

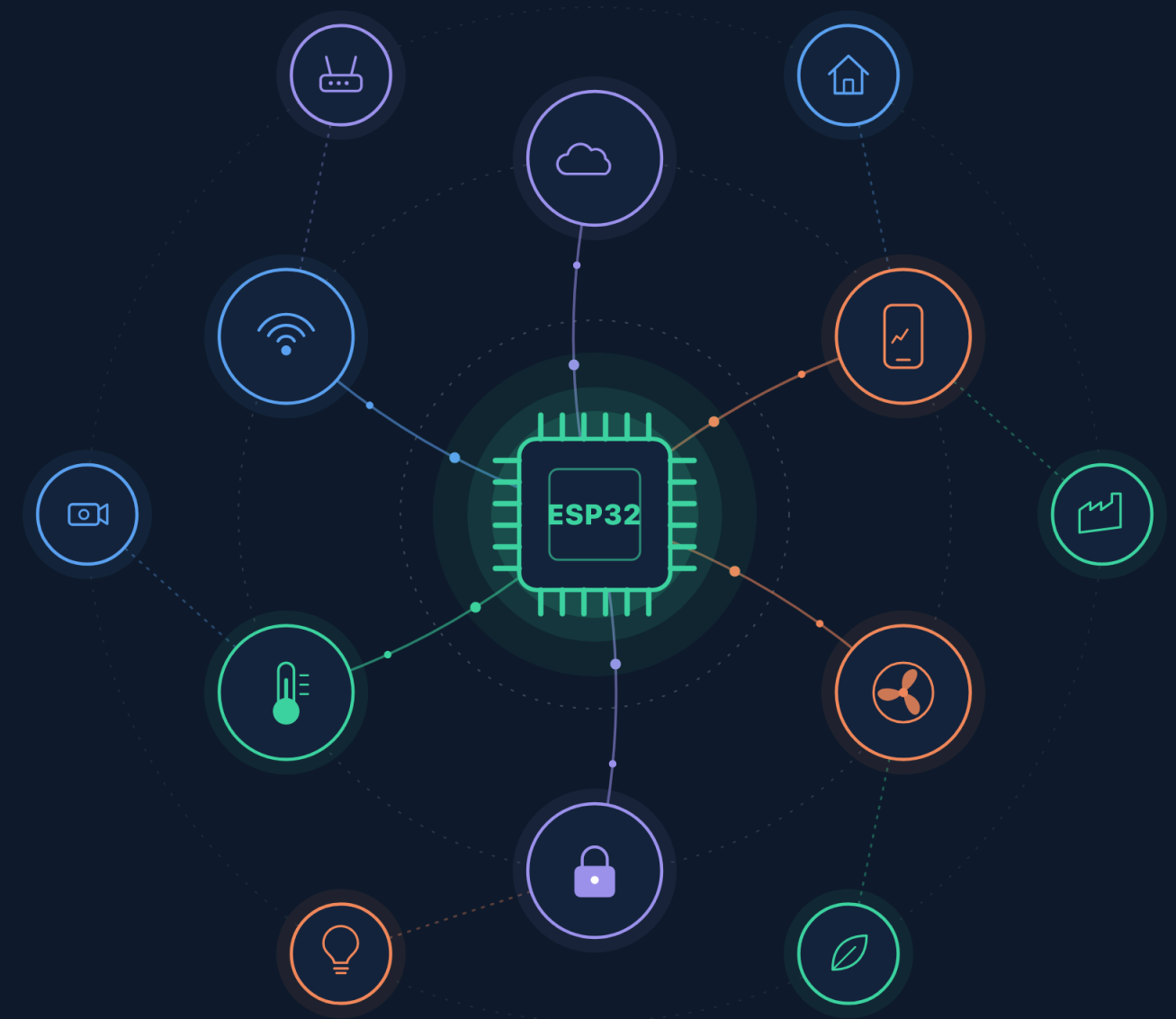
MODULE 1 • 45 MINUTES

What Is the Internet of Things?

Where it came from, what makes a device an IoT device, and where you meet it every day.

Mehdi Lotfinejad

Introduction to the Internet of Things



THE ROAD AHEAD

Eight modules, one smart device

PART 1 — DEVICES AND DATA

01

What is IoT?

You are here

02

How it's built

Lecture

03

Sensors and actuators

Labs 1–2

04

Getting online

Lab 3

PART 2 — CLOUD, AUTOMATION AND SECURITY

05

Cloud and dashboards

Lab 4

06

Automation

Lab 5

07

Security

Group activity

08

Real world + capstone

Lab 6

BY THE END OF THIS MODULE

You will be able to...

-
- 01 **Explain** what the Internet of Things is, in your own words

 - 02 **Describe** how the internet grew from computers to everyday objects

 - 03 **List** the characteristics that make something an IoT device

 - 04 **Explain** why IoT is growing so fast

 - 05 **Give examples** from six areas: homes, cities, industry, health, farms, transport

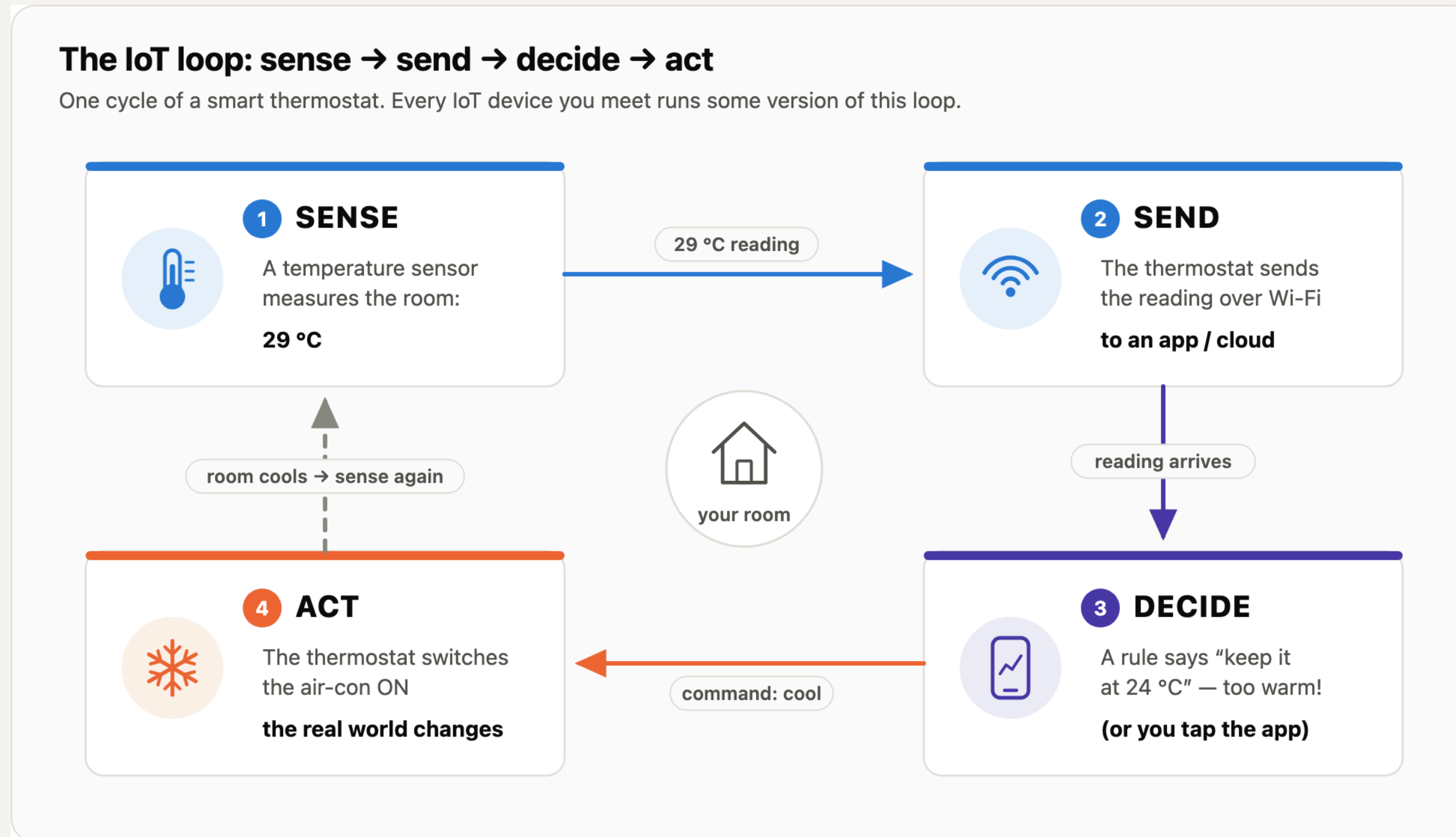
A SIMPLE DEFINITION

The Internet of Things is the network of physical objects that sense the world, share what they sense, and act on it.

A “thing” can be almost anything: a light bulb, a water pump, a streetlight, a heart monitor, or a cow with a smart collar.

THE IOT LOOP

Sense, send, decide, act





A REAL DEVICE

A smart thermostat

- **Sense:** a temperature sensor inside
- **Send:** Wi-Fi to an app
- **Decide:** a rule, or you
- **Act:** switches heating or cooling

PAUSE AND CHECK

A smart doorbell has a camera and a button. Someone presses it, you get a video call, and you unlock the door from your phone. Can you find the four steps?

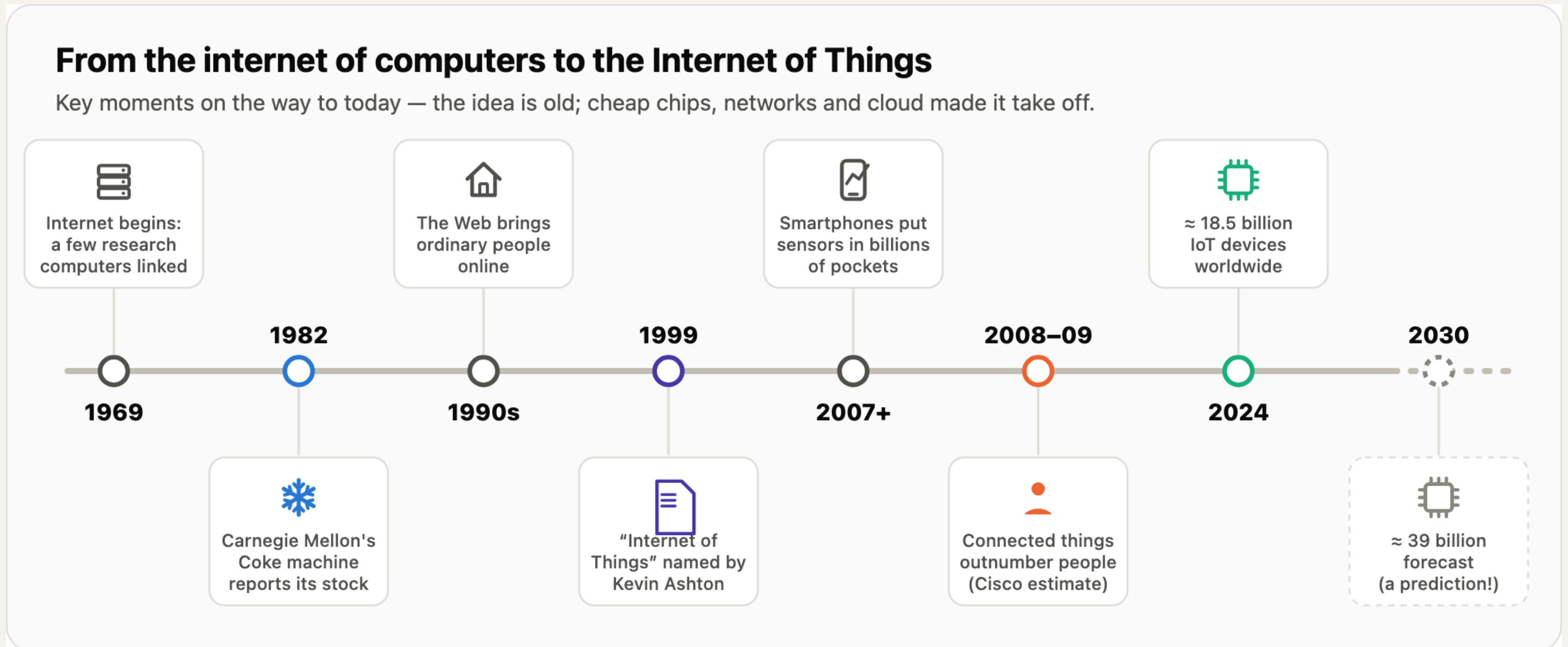
Sense → send → decide → act. Talk to your neighbour for one minute.

Part A

How we got here

From an internet of computers to an Internet of Things.

From a few computers to billions of things





1999 · A NAME IS BORN

Kevin Ashton names the idea

Working on radio tags (RFID) for products, he asked: what if things could report on themselves, instead of people typing in the data?

WHAT MAKES A DEVICE AN IOT DEVICE?

Eight shared characteristics

Characteristic	What it means	Example
Sensing	Measures something in the physical world	A smart meter measures electricity
Connectivity	Sends and receives data over a network	The meter reports to the power company
Unique identity	Each device can be told apart	Each meter has its own ID
Automation	Processes data and acts by itself	A smart plug turns off a heater
Ability to act	Changes the physical world	A smart valve shuts off a leak
Scale	Often huge numbers of devices	A city’s thousands of streetlights
Limited resources	Little power, memory and processing	A soil sensor on one battery for years
Variety	Many makers, still working together	Lights and locks from different brands

If you remember only two: it **senses or acts** on the physical world, and it is **connected**.

Part B

Why now?

The Coke machine was 1982. So why has IoT exploded only in the last fifteen years?

FOUR THINGS GOT CHEAP AT ONCE

Why IoT is growing so fast



Sensors

Smartphones created a huge market for tiny sensors, so prices fell for everyone.



Chips

A microcontroller with Wi-Fi built in, like the ESP32, now costs very little.



Connectivity

Wi-Fi and mobile networks are almost everywhere, even in remote fields.



The cloud

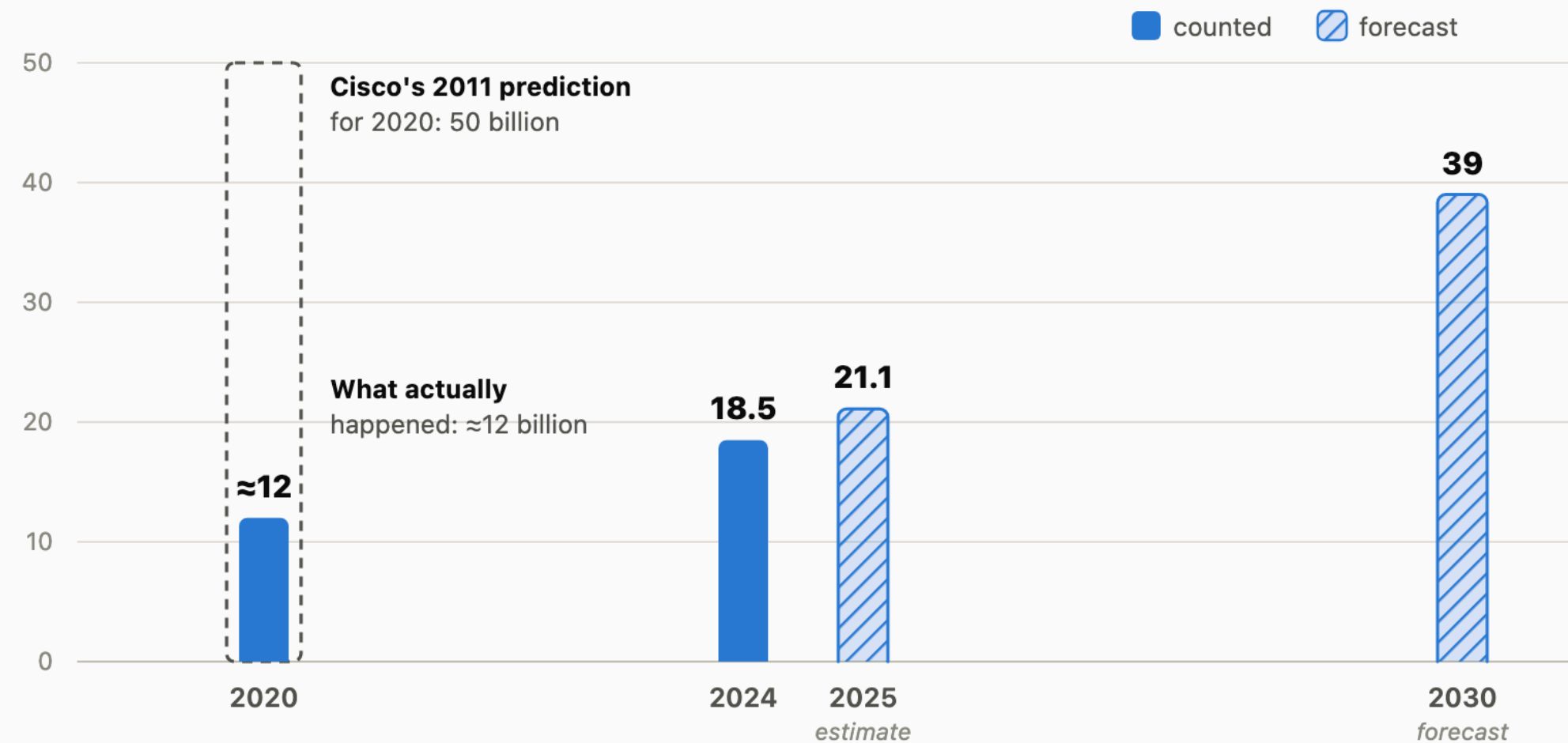
Companies rent storage and computing by the minute, instead of buying servers.

HOW BIG IS IT?

Billions of connected devices

Connected IoT devices worldwide (billions)

Growing fast — but slower than some early predictions. Striped bars are estimates or forecasts, not final counts.



Sources: IoT Analytics, State of IoT 2025 (2024–2030); Cisco IBSG, 2011 (prediction).

18.5 billion IoT devices in 2024

39 billion forecast for 2030

Most connect by **Wi-Fi**, then Bluetooth and cellular

Source: IoT Analytics. Dashed outline: Cisco's 2011 forecast for 2020.

TREAT FORECASTS WITH HEALTHY DOUBT

In 2011, Cisco predicted 50 billion devices by 2020. The real number was about 12 billion.

When you read a big number, ask: who made it, and how did they count?

WHAT'S HAPPENING NOW

Five trends shaping IoT



Smarter edge

Devices decide for themselves; some run small AI models (AIoT).



Matter

A smart-home standard so brands work together.



New networks

LoRaWAN and NB-IoT reach sensors for years on a battery.



Security laws

New rules ban weak default passwords.



Sustainability

Saving energy and water, like dimming empty streets.

Part C

Where is IoT used?

Six application areas — and some examples you probably know.

SIX APPLICATION AREAS

IoT is almost everywhere



Smart homes

Comfort, safety and saving energy.



Smart cities

Streetlights, parking, traffic and air quality.



Industry (IIoT)

Fix machines before they break.



Healthcare

Wearables and patients monitored at home.



Agriculture

Water only where and when crops need it.



Transport

Know where things are and how they are doing.



SMART CITIES

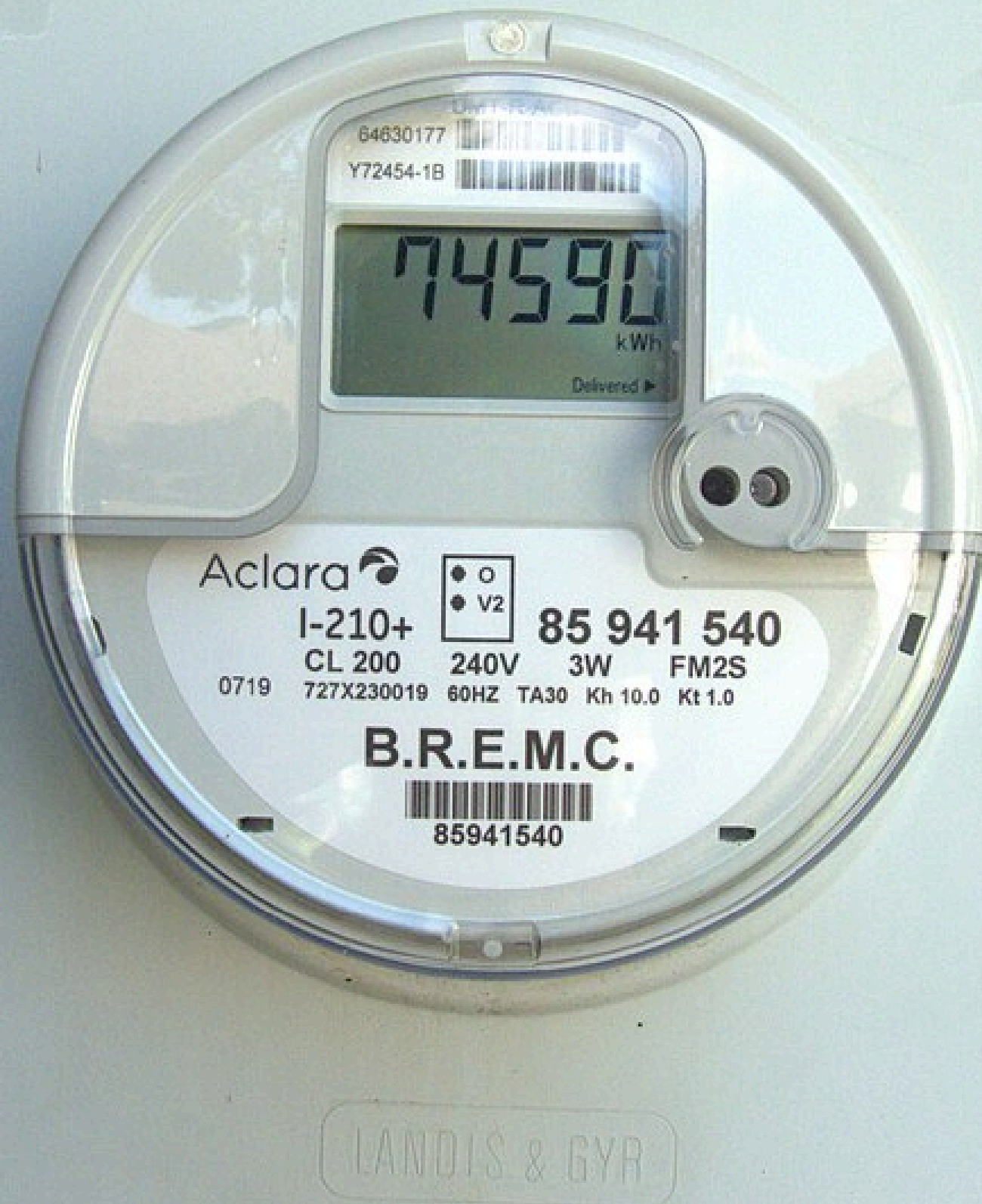
Streetlights that report in

- Los Angeles connected tens of thousands of LED streetlights
- About **64% less energy** than the old lights
- Broken lights report themselves straight away
- Putrajaya runs its own LoRaWAN network for services like smart parking

AN EXAMPLE YOU PROBABLY KNOW

Smart electricity meters

- **Sense** electricity use
- **Send** readings to TNB automatically
- **Decide** the bill, or where a power cut is
- **Act**: crews fix faults; customers save energy
- 5.6 million+ TNB customers already have one



COMMON MISCONCEPTIONS

Four ideas that trip people up

✗ Myth *IoT means controlling things from your phone.*

✓ Fact Most IoT data never reaches a phone. Other computers use it.

✗ Myth *IoT is just gadgets for the home.*

✓ Fact Industry, cities, farms, health and logistics are a huge share.

✗ Myth *An IoT device must be powerful.*

✓ Fact Most are tiny, cheap and low-power — on purpose.

✗ Myth *IoT is a single technology.*

✓ Fact It combines sensors, chips, networks, cloud, data and security.

Where did you meet IoT today?

- 1 List every connected device you used or passed since you woke up.
- 2 For each one, find the loop: sense → send → decide → act.
- 3 Sort them into the six areas: home, city, industry, health, farm, transport.
- 4 Pick one and ask: what data does it collect about me, and who can see it?

KEY TAKEAWAYS

What to remember

- ✓ IoT = physical objects that **sense, share and act** on data
- ✓ Every IoT device follows the loop: **sense → send → decide → act**
- ✓ Named in **1999**; things outnumbered people around **2008–09**
- ✓ Growing fast because sensors, chips, networks and cloud got **cheap**
- ✓ Six areas: **homes, cities, industry, health, farms, transport**

Quick quiz

Q1 A normal scale shows the weight on its screen. A smart scale sends it to an app. Which one is IoT — and why?

Q2 Apply the four steps to an automatic garden sprinkler.

Q3 Give two reasons why IoT has grown so quickly.

Q4 Which area? (a) an overheating factory motor (b) a GPS tracker on a lorry (c) a parking sensor

UP NEXT • MODULE 2

How an IoT System Is Built

We open the “black box”: the five layers of every IoT system, and the journey of one reading from a sensor to a dashboard.



Mehdi Lotfinejad

Book, labs and slides: github.com/datatweets/intro-to-iot-workshop